Jeorge Joshi

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AI/ML engineer with production experience spanning NLP (semantic search, RAG, LLM serving), computer vision (object detection), and cloud-deployed backend infrastructure (AWS, Docker, Kubernetes) at scale. Currently researching large-scale representation learning at UT Arlington.

Education

University of Texas at Arlington — *Expected May 2027*
MS, Computer Science

Tribhuvan University — *July 2024*
BS in Computer Engineering

Experience

Graduate Research Mentor — *Arlington, Texas*
NSF REU Site, University of Texas at Arlington — *Summer 2026 – Present*

- Co-led intensive mentorship for **10** NSF REU undergraduates across **3** animal communication projects (predator-induced corvid/chickadee calls, dog identification, cat breed variation). Established daily code review cycles, **2** weekly team meetings, and all-hands syncs.
- Supervised ML analysis pipelines (segmentation, clustering, embedding evaluation); mentees completed research on vocalization dynamics and individual identification.
- **1** mentee-led demo paper under review at AAAI 2027; **2** mentee-led full papers targeted for ICASSP 2027 submission.

Research Assistant — *Arlington, Texas*
ACL2 Lab, University of Texas at Arlington — *Jan 2026 – Present*

- Performed embedding and unsupervised clustering on a **900K+** sample large-scale acoustic dataset; built segmentation/isolation pipelines (**96–98% F1** in-domain, **68%** zero-shot) scaled to **110K+** files on a multi-GPU cluster.
- Designed a VQ-based tokenization framework revealing domain-mismatch in self-supervised audio models: domain-native tokens achieve **96–98%** F1 while cross-domain transfer collapses to **68%** F1. Exposes fundamental SOTA limitations for non-speech audio—under review at AAAI 2027.
- Evaluated SOTA self-supervised embedding models (HuBERT, AVES) across human and non-human audio domains, quantifying transfer performance and domain-gap failure modes.

AI Engineer — *Bhaktapur, Kathmandu*
DL.surf — *Aug 2024 – July 2025*

- Built semantic search and reranking serving **150K+** daily FastAPI requests with MiniLM, Milvus, Cohere Reranker, and Redis; reduced latency **∼60%**.
- Developed and INT8-quantized multimodal content classifiers (PyTorch → ONNX → TFLite), meeting latency goals on CPU-only infrastructure.
- Designed PostgreSQL schemas, RBAC, connection pooling, indexing, stored procedures, and product-facing REST APIs.
- Dockerized services and owned CI/CD and production deployment with GitHub Actions, Jenkins, Prometheus, and Grafana.

Machine Learning Intern — *Kathmandu, Nepal*
Treeleaf Technologies — *May 2023 – Jul 2023*

- Developed and optimized YOLOv7-based computer vision models for object detection; collaborated on team deployment of ML models to production with GradCam for Explainable AI

Projects

AniSpeakArchive – Animal Communication Literature Search Engine

- Built a corpus-first, resumable 4-stage pipeline (collect → classify → verify/annotate → index) that screened a **410M-record** OpenAlex snapshot and produced a curated corpus of **49,743** animal-communication papers using fine-tuned SciBERT with multi-GPU streaming inference.
- Used Qwen2.5-14B served with vLLM for evidence-grounded metadata extraction and species normalization; achieved **0.963 recall** in screening and enabled faceted filtering across species, modality, research context, and computational method.
- Built hybrid retrieval with BM25, BGE-M3 dense embeddings, reciprocal-rank fusion, and cross-encoder reranking in LanceDB; deployed a Flask app with grounded RAG search, trend analytics, OTP authentication, saved-search alerts, Docker, and Gunicorn.

AI Job Assist – Multi-Agent Job Search and Application Assistant - Link

- Built a multi-agent job-search workflow for candidate onboarding, job collection, fit ranking, tailored resume and cover-letter generation, and application tracking.
- Implemented a drafter–reviewer architecture where an independent reviewer subagent audits generated materials for

factual grounding, requirement coverage, unsupported claims, and keyword relevance before revision and final verification.

- Developed a Python CLI for public Greenhouse and Lever job boards using Pydantic schemas to normalize, deduplicate, and persist postings; added structured candidate profiles, job evaluations, eligibility gates, and ATS-focused LaTeX document verification.

Technologies

Languages: Python, SQL, C++, JavaScript
ML/DL, NLP & CV: PyTorch, TensorFlow, HuggingFace, scikit-learn, Keras, OpenCV, YOLO
GenAI, LLM & Agents: vLLM, Retrieval-Augmented Generation (RAG), Claude Code, Multi-Agent Workflows
Cloud & Deployment: AWS (EC2, ECR, S3), Docker, Kubernetes, GitHub Actions, MLflow, Prometheus, Grafana
Engineering & Automation: Git, LaTeX, REST APIs, CLI development
Data: PostgreSQL, Milvus, LanceDB, Kafka, MongoDB